



University of Windsor

Department of Civil and Environmental Engineering

CIVL8360 - Earthquake Engineering

Term Project – Task 5

Due: Nov 26, 2025

Instructor: Dr. N. Van Engelen

Student Name: Mohammad Ali Azimi

Student ID: 110215457

Contents

1. Executive Summary	3
2. Introduction.....	4
2.1 Background.....	4
2.2 Problem Statement	4
2.3 Objectives	4
3. Theoretical Background.....	5
3.1 Structural Dynamics Fundamentals	5
3.2 Fluid Viscous Dampers (FVDs).....	6
4. Methodology	9
4.1 Structural Model	9
4.2 Damper Configurations.....	10
4.3 Ground Motion Records	11
4.4 Analysis Procedure.....	11
5. Analytical Results	12
5.1 Storey Displacement	12
5.2 Storey Drift	13
5.3 Base Shear.....	13
5.4 Peak Floor Acceleration (PFA)	14
5.5 Performance Ranking.....	15
6. Discussion.....	15
7. Conclusion	16
8. References.....	17
9. Appendices.....	17

1. Executive Summary

This report studies how linear fluid viscous dampers (FVDs) can improve the seismic performance of a three-storey reinforced concrete (RC) frame. A simple 2D model was created in ETABS and was analyzed under the Loma Prieta (1989) and Northridge (1994) earthquakes. Three total damping values (600, 1200, 1800 kN·s/m) were tested in two layouts: one damper at the first storey and three smaller dampers distributed across all storeys. A baseline case without dampers was also included.

In my analyses, all cases with dampers showed smaller displacement, drift, base shear, and floor acceleration. This trend is also reported in previous work on supplemental damping (Goel, 1998). The distributed layout performed better than the single-damper layout in most cases, especially for drift control. The best overall improvement occurred in the case with $C = 1800$ kN·s/m and three distributed dampers.

2. Introduction

2.1 Background

During an earthquake, buildings experience lateral motion that produces displacement, drift, base shear, and acceleration. In RC moment-resisting frames, storey drift is especially important because large drift can damage joints, create cracks, and affect non-structural components. Traditional seismic design increases stiffness and strength, but this can also increase internal forces. For this reason, many modern systems rely on added damping devices that take energy out of the structure without making it stiffer.

Fluid viscous dampers (FVDs) are widely used for this purpose because they generate force based on velocity and provide effective energy dissipation during strong shaking. Several studies have shown that FVDs reduce drift and acceleration in RC and steel frames (Goel, 1998).

2.2 Problem Statement

While FVDs can improve seismic behavior, two important design choices must be made:

1. How much damping should be added?
2. Where should the dampers be placed?

If the total damping is too small, the improvement is limited; if too large, damper forces may become excessive (Qian et al., 2020). Placing a single damper at one level may not control upper-storey drift, while distributing dampers can provide more balanced vibration control (Pettinga et al., 2013). In this project, I used a simple numerical model to study these two points in a clear way.

2.3 Objectives

The objectives of this project are:

- Evaluate how linear FVDs affect the seismic response of a three-storey RC frame.
- Compare three total damping values: 600, 1200, and 1800 kN·s/m.
- Compare two layouts: one damper at the first storey and three dampers across the height.
- Examine displacement, drift, base shear, and acceleration.
- Compare all configurations with the baseline case.
- Identify the most effective configuration under two earthquakes.

3. Theoretical Background

The theoretical background explains the main principles of structural dynamics and the behavior of fluid viscous dampers (FVDs). These concepts help clarify why dampers reduce earthquake effects and why damper amount and placement strongly influence performance.

3.1 Structural Dynamics Fundamentals

Earthquake ground motion causes the building's base to move horizontally while the upper floors lag. The motion can be represented using the SDOF equation:

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = -m\ddot{u}_g(t)$$

Where:

- m = mass
- c = damping
- k = stiffness
- $u(t)$ = displacement
- $\ddot{u}_g(t)$ = ground acceleration

The terms mean:

- $ku(t)$ = restoring force from stiffness
- $c\dot{u}(t)$ = damping force
- $-m\ddot{u}_g(t)$ = effect of ground motion

For real buildings, which have many floors, the system becomes a multi-degree-of-freedom (MDOF) model. ETABS automatically builds the mass, stiffness, and damping matrices and solves the equations numerically.

Important Seismic Response Measures

For multi-storey buildings, this becomes a system of equations. Important response quantities include:

- storey displacement (movement of each floor)
- storey drift (relative movement between floors)
- base shear (total horizontal force at base)

- floor acceleration (important for non-structural elements)

3.2 Fluid Viscous Dampers (FVDs)

Fluid viscous dampers are mechanical devices that reduce earthquake demands by converting vibration energy into heat through controlled fluid motion. They are commonly used in buildings and bridges because they work in both small and large earthquakes and do not change the stiffness of the structure.

FVDs act only in the axial direction along the line between their two end connections. They do *not* resist bending, rotation, or shear. This means the designer must place them in a way that the relative motion of the two nodes creates a strong axial compression in the damper.

Components of an FVD

A typical FVD contains:

- A steel cylinder filled with a viscous silicone-based fluid.
- A piston rod connected to one structural joint.
- A piston head with orifices, where the fluid is forced to pass through.
- End connections that link the two ends of the damper to the frame.
- High-pressure seals that prevent fluid leakage and allow long-term use.

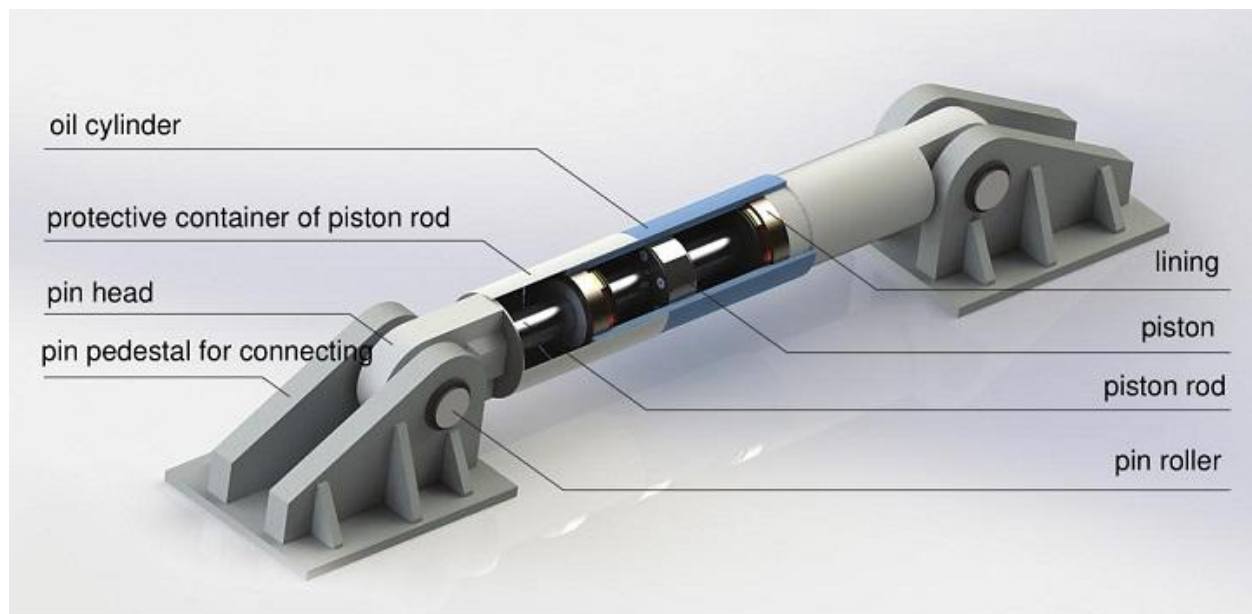


Figure 1 – Damper Structure

How FVDs Work

During an earthquake, two ends of the damper move relative to each other. This forces the piston to move, and the fluid is pushed through small openings. The resistance to fluid flow creates a damping force.

This force is described by the velocity relationship:

$$F = Cv^\alpha$$

Where:

-
- F = damper force
- C = damping coefficient
- v = relative velocity
- α = velocity exponent

In this project, $\alpha = 1$, creating a linear viscous damper:

$$F = Cv$$

Earlier work also shows that linear FVDs can reduce deformation demands in RC frames (Lamprea Pineda et al., 2019).



Figure 2 – V-shape FVD installed in building



Figure 3 – FVD installed in bridge



Figure 4 – FVD installed in building

Energy Dissipation

An FVD removes energy by converting kinetic energy into heat inside the fluid. This heat stays inside the damper and does not affect the structure.

Energy dissipation reduces:

- Storey drift
- Storey displacement
- Peak acceleration
- Vibration duration

Effect of Damping Amount (C Value)

The damping coefficient **C** controls the damper strength.

- Higher $C \rightarrow$ larger velocity-dependent force
- Larger force $\rightarrow$ more energy removed
- More energy removed $\rightarrow$ lower drift and displacement

Several studies have shown that increasing viscous damping usually lowers seismic demands, especially in stronger earthquakes, especially under strong motion (e.g., Northridge-type events) (Qian et al., 2020).

Effect of Damper Layout

Where the dampers are placed is just as important as how strong they are.

- **Single-Damper Layout**
 - All damping placed at one storey
 - Strong improvement at that floor
 - Weaker improvement at higher floors
 - Does not control higher modes well
- **Distributed-Damper Layout (3 Small Dampers)**
 - One damper per storey
 - More uniform control
 - Strong drift reduction at all floors
 - Better performance under strong, pulse-like earthquakes

Research agrees that distributed damping provides better overall seismic control (Pettinga et al., 2013).

4. Methodology

This section explains how the three-storey RC frame was modeled, how the dampers were assigned, how the earthquake records were used, and how the nonlinear time-history analyses were performed in ETABS.

4.1 Structural Model

The structure used in this project is a three-storey, three-bay reinforced concrete (RC) moment frame modeled in ETABS.

Main features of the model

- Three bays in the horizontal direction
- Three equal storey heights
- RC beams and columns with typical rectangular sections

- Material properties from standard concrete (ETABS default)
- Fixed supports at the base
- Rigid diaphragm constraint at each floor
- Lumped masses at floor levels using ETABS automatic mass source

4.2 Damper Configurations

Fluid viscous dampers (FVDs) are modeled using nonlinear link elements in ETABS. Only the axial degree of freedom is activated. The force–velocity law is linear:

$$F = Cv$$

Three total damping coefficients are used:

- $C = 600 \text{ kN}\cdot\text{s/m}$
- $C = 1200 \text{ kN}\cdot\text{s/m}$
- $C = 1800 \text{ kN}\cdot\text{s/m}$

Two layouts were considered:

- One large damper at the first storey.
- Three smaller dampers distributed across the height.

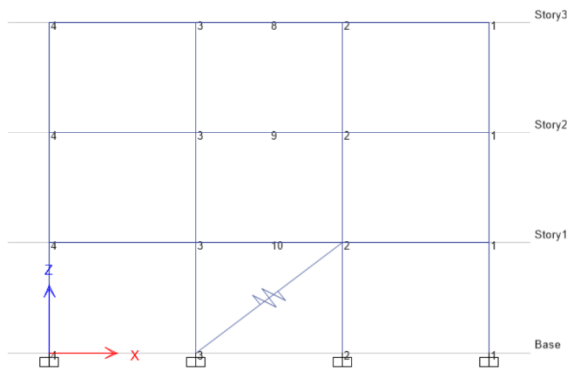


Figure 5- 1 Big Damper

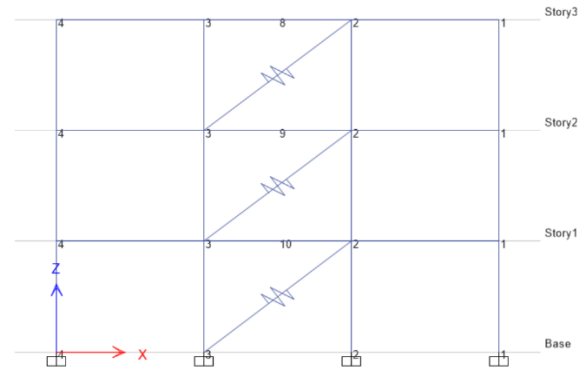


Figure 6 – 3 Small Dampers

Seven analysis cases:

- Baseline (No damper)
- C600-1 Damper, C1200-1 Damper, C1800-1 Damper

- C600-3 Dampers, C1200-3 Dampers, C1800-3 Dampers

These cases make it easier to compare how the amount and placement of damping affect the frame.

4.3 Ground Motion Records

Two real ground motions were applied:

- Loma Prieta (1989): far-field motion
- Northridge (1994): near-fault motion with velocity pulses

Using both records follows recommendations in damping studies where near-fault pulses significantly influence damper response (Lamprea Pineda et al., 2019).

4.4 Analysis Procedure

Steps:

1. Build RC frame in ETABS
2. Define materials and sections
3. Assign mass
4. Define linear viscous damper
5. Create all cases (baseline + six damper cases)
6. Import ground motions
7. Run nonlinear time-history analysis
8. Extract peak response values
9. Export to Excel
10. Plot in MATLAB

5. Analytical Results

This section presents the peak response values obtained from ETABS for all seven analysis cases and both earthquakes. The goal is to compare how damper amount and damper placement influence the seismic performance of the three-storey RC frame.

All results were taken directly from the ETABS output and the plots prepared for the project.

5.1 Storey Displacement

In every damper case, the displacements were lower, and the higher C values generally performed better. Distributed damping improved performance at upper floors, which aligns with design approaches for distributing supplemental damping described by Pettinga et al. (2013).

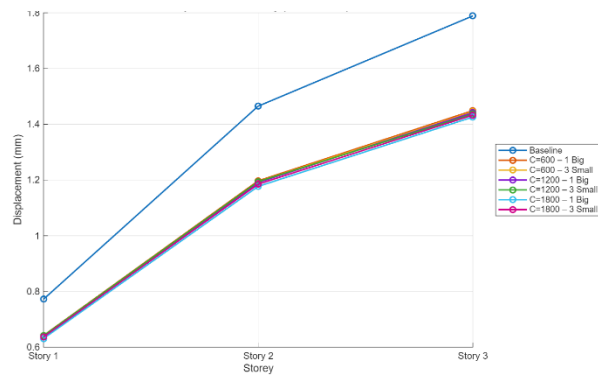


Figure 7 – Displacement vs Storey (Loma Prieta)

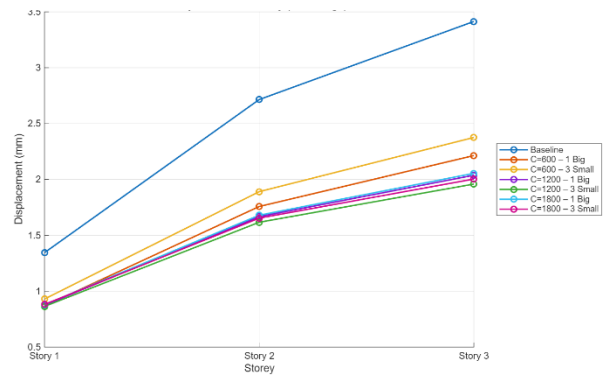


Figure 8 – Displacement vs Storey (Northridge)

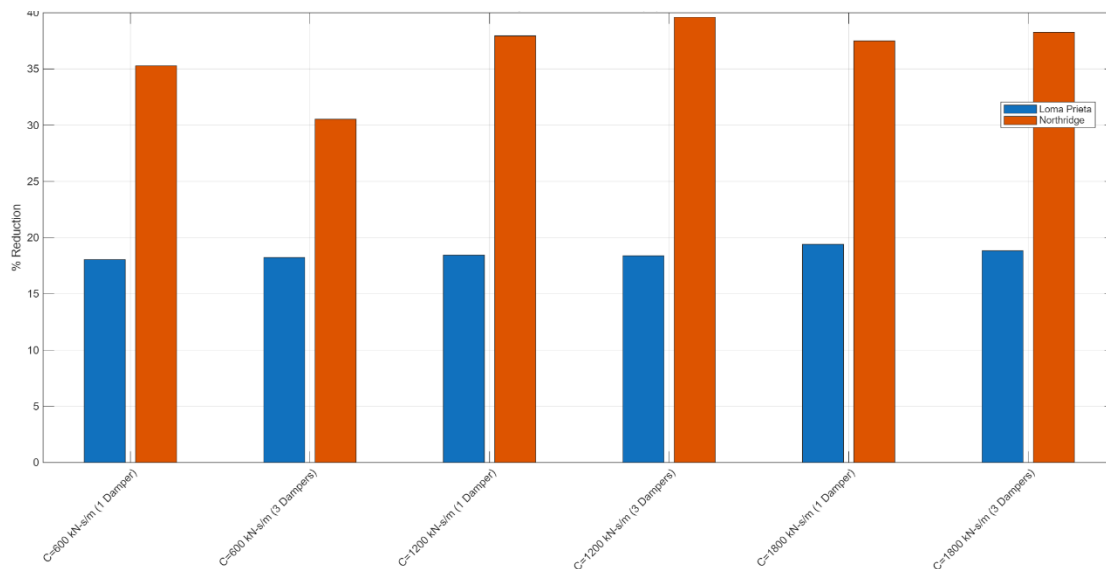


Figure 9 – Displacement Reduction %

5.2 Storey Drift

Drift also dropped in every case, and the reductions were larger under the Northridge record because of its high velocities, because velocity pulses generate higher damper forces in linear devices (Qian et al., 2020). The distributed layout achieved the best results.

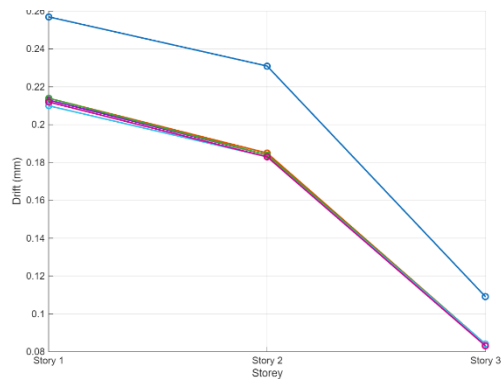


Figure 11 – Drift vs Storey (Loma Prieta)

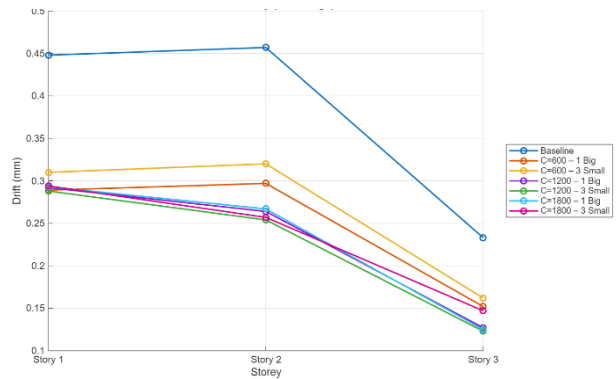


Figure 10 – Drift vs Storey (Northridge)

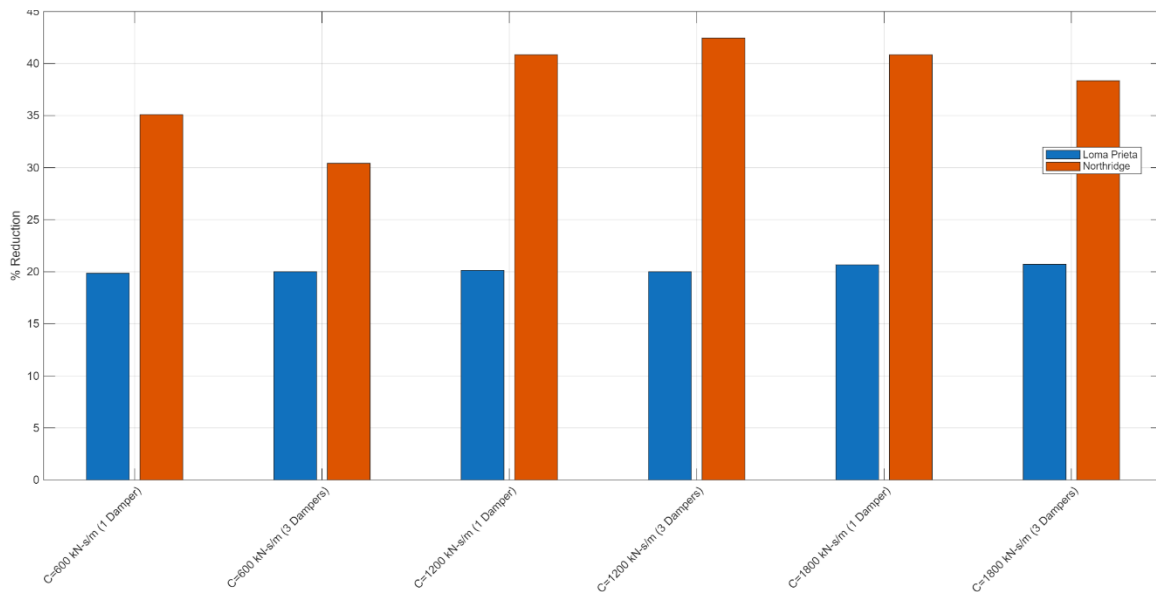


Figure 12 – Drift Reduction (%)

5.3 Base Shear

Base shear also went down in every case, but the drop was smaller compared with drift and displacement. Similar findings were reported by Goel, (1998).

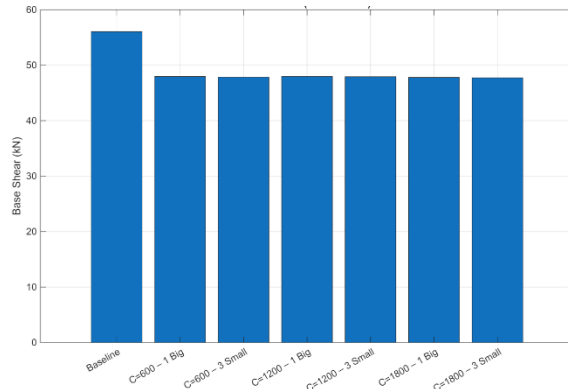


Figure 13 – Base Shear (Loma Prieta)

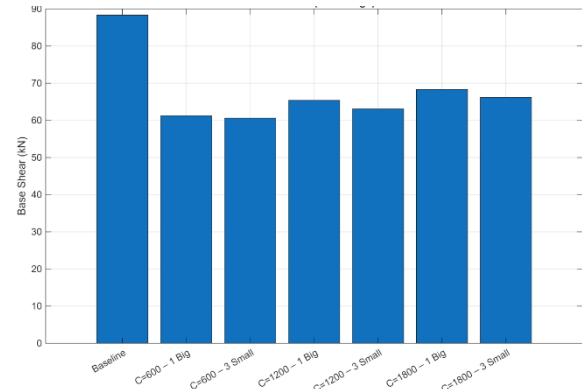


Figure 14 – Base Shear (Northridge)

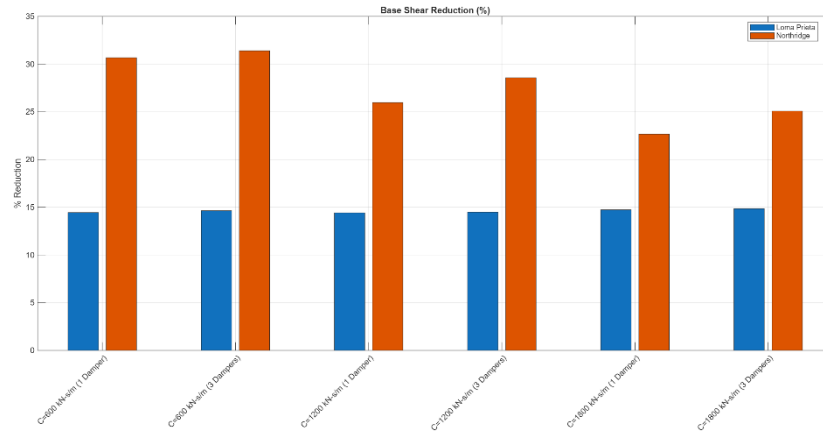


Figure 15 – Base Shear Reduction (%)

5.4 Peak Floor Acceleration (PFA)

Accelerations decreased for all damper configurations, with larger improvements at higher damping values. The distributed layout showed better control of floor accelerations, especially at higher damping values, consistent with observations in Lamprea Pineda et al, (2019).

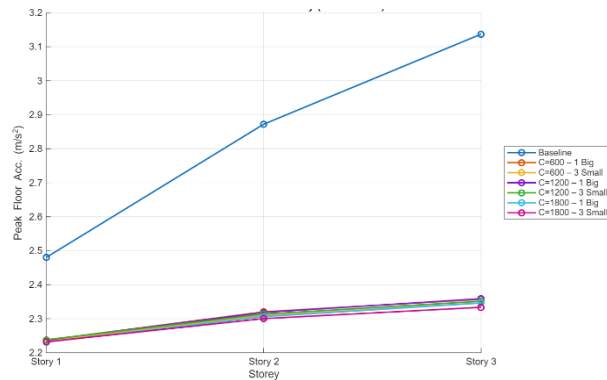


Figure 16 – PFA vs Storey (Loma Prieta)

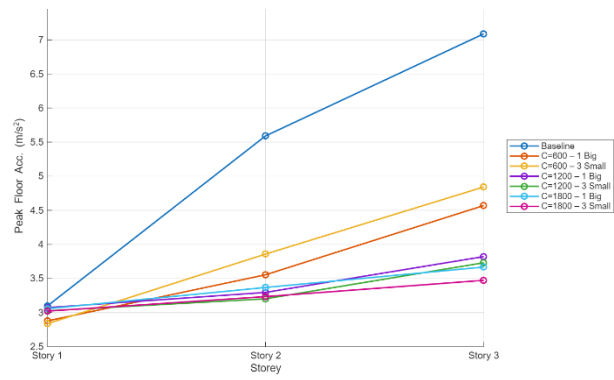


Figure 17 – PFA vs Storey (Northridge)

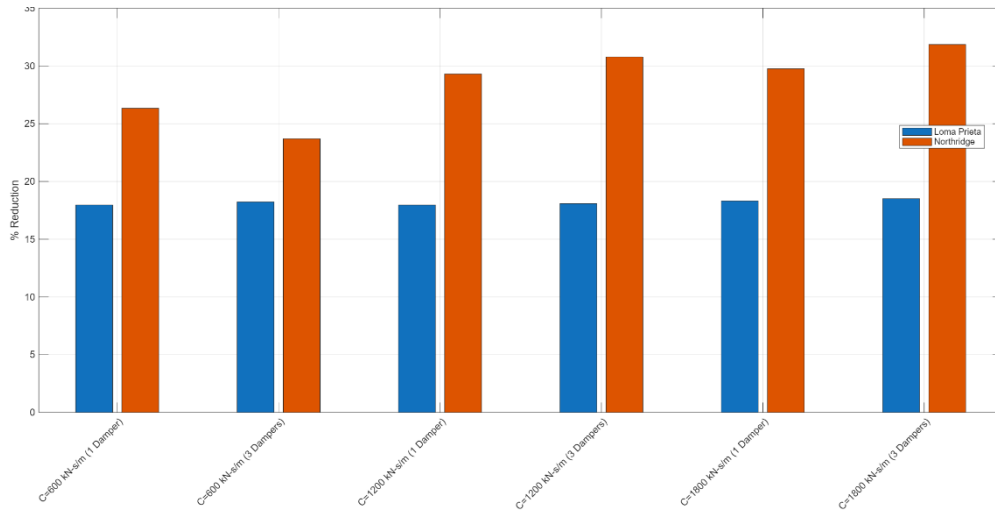


Figure 18 – PFA Reduction (%)

Reasons for Larger Reductions Under the Northridge Record

For Loma Prieta, the average reductions in drift, displacement, base shear and acceleration are around 18 - 20%, which is consistent with the typical values reported in many studies for code-designed buildings. However, for Northridge, the reductions are much larger, often in the range of 30 - 40% and up to about 50% for some response measures. This is not an error; it is mainly due to the strong near-fault velocity pulses in the Northridge record and the velocity-dependent nature of fluid viscous dampers. Because the damper force is proportional to velocity, FVDs become much more effective under this type of ground motion. In addition, the model used in this project is a simple three-storey frame with idealized linear behavior and relatively high total damping (up to $C = 1800 \text{ kN}\cdot\text{s/m}$) distributed at all storeys, which further amplifies the apparent percentage reduction compared to more complex real buildings.

5.5 Performance Ranking

Across all response parameters and both earthquakes, the best configuration was $C = 1800 \text{ kN}\cdot\text{s/m}$ with three distributed dampers, which is like what many studies recommend about spreading damping across different storeys (Lamprea Pineda et al., 2019).

6. Discussion

The results clearly show that adding viscous damping helps lower the seismic demands, especially drift and displacement. Distributed damping performed better because it influences both first mode and higher-mode behavior, which is consistent with trends reported in previous research (Pettinga et al., 2013). The Northridge record produced greater percentage reductions due to its strong velocity pulses, which activate linear dampers more effectively (Qian et al., 2020).

One practical drawback of $C = 1800$ is that the damper forces might be high, but this was not checked in

this project. The study uses a simple model, so real buildings may show more complex behavior, but the general trends remain valid.

7. Conclusion

This project examined how linear FVDs influence the seismic response of a three-storey RC frame. All damper configurations reduced displacement, drift, base shear, and acceleration. Higher damping values provided better performance, and the distributed layout produced the most uniform reductions. The best configuration was **$C = 1800 \text{ kN}\cdot\text{s/m}$ with three distributed dampers**, that also agrees with findings from other studies that support using distributed damping in multi-storey frames (Pettinga et al., 2013).

8. References

- Dong, S., Pan, W., Ye, L., & Wang, J. (2025). Design of displacement-based viscous damper damping structures. *Scientific Reports*, 15(1), Article 11742.
- Goel, R. K. (1998). Effects of supplemental viscous damping on seismic response of asymmetric-plan systems. *Earthquake Engineering and Structural Dynamics*, 27(2), 125–141.
- Lamprea Pineda, A. C., Garzón Amortegui, L. F., & Chesi, C. (2019). Use of viscous fluid dampers for the improvement of the seismic response of RC structures. *Vibroengineering PROCEDIA*, 23, 87–92.
- Pettinga, J. D., Oliver, S., & Kelly, T. E. (2013, February 21–22). *A design office approach to supplemental damping using fluid viscous dampers* [Paper presentation]. Steel Innovations Conference 2013, Christchurch, New Zealand.
- Qian, X., Chopra, A. K., & McKenna, F. (2020). *Modeling viscous damping in nonlinear response history analysis of steel moment-frame buildings: Design-plus ground motions* (PEER Report No. 2020/01). Pacific Earthquake Engineering Research Center.

9. Appendices

Appendix A - ETABS Model Information

A.1 Geometry

- 3-storey, 3-bays reinforced concrete moment frame
- Equal storey height
- 3 horizontal spans
- Fixed base supports
- Rigid diaphragm at each floor
- Lumped mass defined at each storey level

A.2 Material Properties

- Concrete material (ETABS default)
- Reinforcing steel (ETABS default)

A.3 Section Properties

- RC columns: rectangular section
- RC beams: rectangular section

A.4 Mass Source

- Self-weight
- Superimposed dead load
- Assigned automatically by ETABS

Appendix B - Ground Motion Records

B.1 Loma Prieta (1989)

- Original accelerogram used
- No scaling applied
- Full duration and time-step preserved

B.2 Northridge (1994)

- Original accelerogram used
- No scaling
- Strong near-fault velocity pulses

Appendix C - Damper Properties and Layouts

C.1 Link Properties

- linear viscous damper
- Force-velocity law: $F = C\dot{v}$
- Velocity exponent: $\alpha = 1$
- Only axial degree of freedom activated (U1)

C.2 Damping Values

Case	Total C (kN·s/m)	Layout
Baseline	0	No dampers
C600-1 Big D	600	1 damper at Storey 1
C1200-1 Big D	1200	1 damper at Storey 1
C1800-1 Big D	1800	1 damper at Storey 1
C600-3 Small D	600	3 dampers (200 each)
C1200-3 Small D	1200	3 dampers (400 each)
C1800-3 Small D	1800	3 dampers (600 each)

Appendix D - Numerical Peak Values

Max Displacement:

		Max Displacement						C=600
		1 Big Damper (mm)			3 Small Dampers (mm)			
		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	
Loma Prieta	Northridge	0.642	1.197	1.449	0.641	1.194	1.445	
		0.868	1.758	2.214	0.93	1.891	2.376	

		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1200
Loma Prieta	Northridge	0.639	1.191	1.442	0.641	1.194	1.437	
		0.875	1.667	2.038	0.864	1.617	1.959	

		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1800
Loma Prieta	Northridge	0.631	1.177	1.426	0.637	1.185	1.433	
		0.88	1.68	2.055	0.882	1.654	2.004	

		Story 1	Story 2	Story 3				Baseline
Loma Prieta	Northridge	0.772	1.465	1.789				
		1.345	2.715	3.414				

		Reduction %						C=600
		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	
Loma Prieta	Northridge	16.84	18.29	19.01	16.97	18.50	19.23	
		35.46	35.25	35.15	30.86	30.35	30.40	

		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1200
Loma Prieta	Northridge	17.23	18.70	19.40	16.97	18.50	19.68	
		34.94	38.60	40.30	35.76	40.44	42.62	

		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1800
Loma Prieta	Northridge	18.26	19.66	20.29	17.49	19.11	19.90	
		34.57	38.12	39.81	34.42	39.08	41.30	

Max Storey Drift:

		Max Storey Drift						
		1 Big Damper (mm)			3 Small Dampers (mm)			
		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=600 kN-s/m
Loma Prieta Northridge		0.214	0.185	0.084	0.214	0.184	0.084	
		0.289	0.297	0.152	0.31	0.32	0.162	

		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1200 kN-s/m
Loma Prieta Northridge		0.213	0.184	0.084	0.214	0.184	0.084	
		0.292	0.264	0.127	0.288	0.254	0.123	

		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1800 kN-s/m
Loma Prieta Northridge		0.21	0.183	0.084	0.212	0.183	0.083	
		0.293	0.267	0.125	0.294	0.257	0.147	

		Story 1	Story 2	Story 3				Baseline
Loma Prieta Northridge		0.257	0.231	0.109				
		0.448	0.457	0.233				

		Reduction %						
		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	
		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	
Loma Prieta Northridge		16.73	19.91	22.94	16.73	20.35	22.94	C=600 kN-s/m
		35.49	35.01	34.76	30.80	29.98	30.47	
		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1200 kN-s/m
		17.12	20.35	22.94	16.73	20.35	22.94	
		34.82	42.23	45.49	35.71	44.42	47.21	
		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1800 kN-s/m
		18.29	20.78	22.94	17.51	20.78	23.85	
		34.60	41.58	46.35	34.38	43.76	36.91	

Base Shear:

Base Shear			
	1 Big Damper (kN)	Base Shear	
Loma Prieta Northridge	61.2443	60.5954	C=600 kN-s/m
	47.9531	47.8438	
Loma Prieta Northridge	65.3908	63.0929	C=1200 kN-s/m
	47.9776	47.9283	
Loma Prieta Northridge	68.3064	66.1782	C=1800 kN-s/m
	47.7954	47.726	
Loma Prieta Northridge	88.3328		Baseline
	56.0519		

Reduction %			
Loma Prieta Northridge	14.45	14.64	C=600 kN-s/m
	30.67	31.40	
	14.41	14.49	C=1200 kN-s/m
	25.97	28.57	
	14.73	14.85	C=1800 kN-s/m
	22.67	25.08	

PFA:

		Peak Floor Acceleration						
		1 Big Damper (mm)			3 Small Dampers (mm)			
		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	
Loma Prieta Northridge		2.237	2.3202	2.3573	2.2346	2.3091	2.3468	C=600 kN-s/m
		2.8769	3.5539	4.5697	2.8353	3.8575	4.8425	

		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1200 kN-s/m
Loma Prieta Northridge		2.2372	2.3186	2.3589	2.2381	2.3136	2.3516	
		3.0717	3.2952	3.8208	3.0254	3.2013	3.7305	

		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1800 kN-s/m
Loma Prieta Northridge		2.2313	2.3061	2.3462	2.2318	2.3003	2.3336	
		3.0561	3.3679	3.6671	3.0205	3.2337	3.4726	

	Story 1	Story 2	Story 3	Baseline
Loma Prieta	2.4801	2.8721	3.1371	
Northridge	3.0948	5.5907	7.0888	

		Reduction (%)						
		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	
Loma Prieta Northridge		9.80	19.22	24.86	9.90	19.60	25.19	C=600 kN-s/m
		7.04	36.43	35.54	8.39	31.00	31.69	

		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1200 kN-s/m
Loma Prieta Northridge		9.79	19.27	24.81	9.76	19.45	25.04	
		0.75	41.06	46.10	2.24	42.74	47.37	

		Story 1	Story 2	Story 3	Story 1	Story 2	Story 3	C=1800 kN-s/m
Loma Prieta Northridge		10.03	19.71	25.21	10.01	19.91	25.61	
		1.25	39.76	48.27	2.40	42.16	51.01	

Appendix E - Plots of Results

